ECA1804 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO5 – CASE STUDY BASED ASSIGNMENT

(1 × 100 = 100 Marks)

Case Study Based Assignment Report

Diagnosis, Repair and 1.6 Tb/s Upgrade of a 60 km DWDM Metro Link: A Budget-First Approach with 400G Si-Photonic Pluggables

Submitted by

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Abstract

A 60 km G.652.D metro link carrying 8 × 10 Gb/s NRZ channels failed acceptance testing: Channel 5 showed a partially closed eye and high BER, and even the healthy Channel 3 had less than the required 3 dB receiver margin. This report works through the field data step by step. The as-built loss is 22.52 dB against a design value of 21.25 dB; the 1.27 dB excess comes from a contaminated or damaged LC/UPC patch-panel connector (event 3, +0.70 dB) and a degraded mechanical splice (event 2, +0.57 dB). With the 1.0 dB dispersion penalty and the extinction-ratio penalty, Channel 3 has only 0.38 dB margin and Channel 5 has −1.75 dB. The link is power-limited, not noise-limited (OSNR = 47 dB). The Channel 5 problem is channel-specific: its 4.5 dB extinction ratio alone predicts Q = 4.66, matching the measured 4.7. The recommended actions are to replace event 3 with an LC/APC connector, re-make event 2 as a fusion splice, raise the launch power to +3 dBm and re-tune or replace the Channel 5 transmitter, which restores 3.65 dB margin. For the 1.6 Tb/s upgrade, Option R (4 × 400G Si-photonic pluggables) is selected: it uses 300 GHz of spectrum (5.33 b/s/Hz), has 21 dB OSNR margin and needs only 37.5 pJ/bit. The final decision is YES – deploy.

1. Stage 1 – Interpreting the Measurement and Performance Requirement

1.1 Instruments and what each validates

Instrument	Quantity measured	What it validates on this link
Optical power meter + stable light source	Absolute power (dBm), insertion loss (dB)	End-to-end loss, launch power (+1 dBm), received power vs −24 dBm sensitivity, individual connector loss (0.5 dB readings)
Optical spectrum analyser (OSA)	Power per channel, OSNR in 0.1 nm, centre wavelength	Channel plan on the 100 GHz grid, booster EDFA OSNR (target 20 dB), power tilt between channels
OTDR	Backscatter vs distance, event loss and reflectance	Fibre attenuation (0.21 dB/km), splice/connector position and loss, location of faults
Eye-diagram / BER tester (DCA + BERT)	Eye opening, ER, jitter, Q-factor, BER	Signal quality per channel against BER 10^{-12} ($Q \geq 7.03$)
Dispersion / PMD analyser	CD (ps/nm), DGD (ps)	Accumulated CD (1020 ps/nm) and PMD against tolerance

1.2 Components of a practical budget

A realistic budget must include more than fibre loss. The **power budget** contains fibre attenuation, connector and splice losses, the 6 dB demultiplexer, power penalties (dispersion, extinction ratio, jitter) and the 3 dB design margin. The **dispersion budget** uses D, link length, bit rate and source spectral width. The **OSNR budget** depends on the booster EDFA noise figure and its input power. For the coherent upgrade, the required OSNR of each transceiver replaces the power sensitivity as the main constraint.

1.3 Eye opening, jitter, ER and BER

The eye diagram is the overlay of all bit transitions. **Vertical opening** reflects noise and the separation of the 1 and 0 levels; it sets the Q-factor, $Q = (I_1 - I_0)/(\sigma_1 + \sigma_0)$. **Horizontal opening** reflects timing jitter (deterministic DJ plus random RJ). A **low extinction ratio** (P_1/P_0) means the 0 level is not dark enough, so part of the received power carries no information; Q falls and a power penalty appears. Since BER = ½ erfc(Q/√2), Q must be at least 7.03 for BER 10^{-12}.

1.4 OTDR limitations

Dynamic range limits how far along the fibre the trace stays above the noise floor. **Event dead zone (EDZ)** is the minimum distance after a reflective event before a second event can be seen as separate. **Attenuation dead zone (ADZ)** is the minimum distance before the loss of a second event can be measured. On this link the two patch-panel connectors are only 9 m apart, so the ADZ, not the dynamic range, is the limiting factor at the far end.

1.5 Standards and component choices

- **ITU-T G.652.D** – standard single-mode fibre with low water peak; attenuation ≤ 0.3 dB/km at 1550 nm, D ≈ 17 ps/(nm·km), PMD_Q ≤ 0.2 ps/√km. The measured 0.21 dB/km and 0.1 ps/√km are within this specification, so the fibre itself does not need re-laying.
- **ITU-T G.694.1** – DWDM frequency grid anchored at 193.1 THz, fixed spacings (100/50/25/12.5 GHz) and flexible grid with 12.5 GHz slot granularity.
- **Fusion splice** (≈ 0.02–0.05 dB, no reflection, permanent) versus **mechanical splice** (0.1–0.5 dB, reflectance −40 to −60 dB when new, gel can degrade).
- **Connector families:** SC (2.5 mm ferrule), LC (1.25 mm, high density, standard in SFP+/QSFP), MPO (12–24 fibre array, for parallel optics) and CS (very small form factor duplex for 400G/800G modules). Polish: UPC (RL ≥ 50 dB) or APC (8° angle, RL ≥ 60 dB).

1.6 Constraints

The fibre pair cannot be re-laid, so all improvements must come from terminal equipment, connectors, splices and amplifier settings. The repaired link must give ≥ 3 dB margin, the upgrade must reach 1.6 Tb/s on the same pair, use as little spectrum as possible, and minimise energy per bit.

Performance Requirement Statement: After repair, the 60 km G.652.D link shall have a total loss no greater than the 21.25 dB design value, give every 10G channel a received power at least 3 dB above −24 dBm after all penalties, show no OTDR event above its allowance, and achieve Q ≥ 7.03 (BER ≤ 10^{-12}) with ER ≥ 8.2 dB on every channel. The upgraded system shall then carry 1.6 Tb/s on the same fibre pair on a G.694.1-compliant grid, with positive OSNR margin and the lowest practical energy per bit.

2. Stage 2 – Link Budget, Dispersion and OSNR

2.1 Design versus as-built loss

$$\textit{Fibre attenuation} = 60\ \text{km} \times 0.21\ \text{dB/km} = 12.60\ \text{dB}$$

Loss component	Design allowance (dB)	As-built (measured) (dB)	Difference (dB)
Fibre (60 km × 0.21)	12.60	12.60	0
Demultiplexer	6.00	6.00	0
Connectors	4 × 0.5 = 2.00	1.20 (event 3) + 3 × 0.5 = 2.70	+0.70
Splices	13 × 0.05 = 0.65	12 × 0.05 + 0.62 (event 2) = 1.22	+0.57
Total	**21.25**	**22.52**	**+1.27**

2.2 Received power and penalties

$$P_{rx} = P_{launch} - L_{total} = +1 - 22.52 = -21.52\ \text{dBm per channel}$$

Dispersion penalty:

$$B \cdot D \cdot L \cdot \sigma\lambda = (10 \times 10^{9})(17 \times 10^{-12})(60)(0.03) = 0.306$$

$$\delta_{CD} = 5\log_{10}[1 + 2\pi(0.306)^{2}] = 5\log_{10}(1.588) = 1.00\ \text{dB}$$

Extinction-ratio penalty: $\delta_{ER} = 10\log_{10}[(r+1)/(r-1)]$, where r is the linear ER.

Channel	ER (dB)	r (linear)	δ_ER (dB)	Effective P_rx (dBm)	Margin vs −24 dBm (dB)	≥ 3 dB?
3	9.0	7.94	1.10	−21.52 − 1.00 − 1.10 = −23.62	+0.38	No
5	4.5	2.82	3.22	−21.52 − 1.00 − 3.22 = −25.75	−1.75	No (fails)

Note that Channel 5 also violates the condition under which the −24 dBm sensitivity is specified (ER ≥ 8.2 dB), so its real sensitivity is worse than the datasheet value.

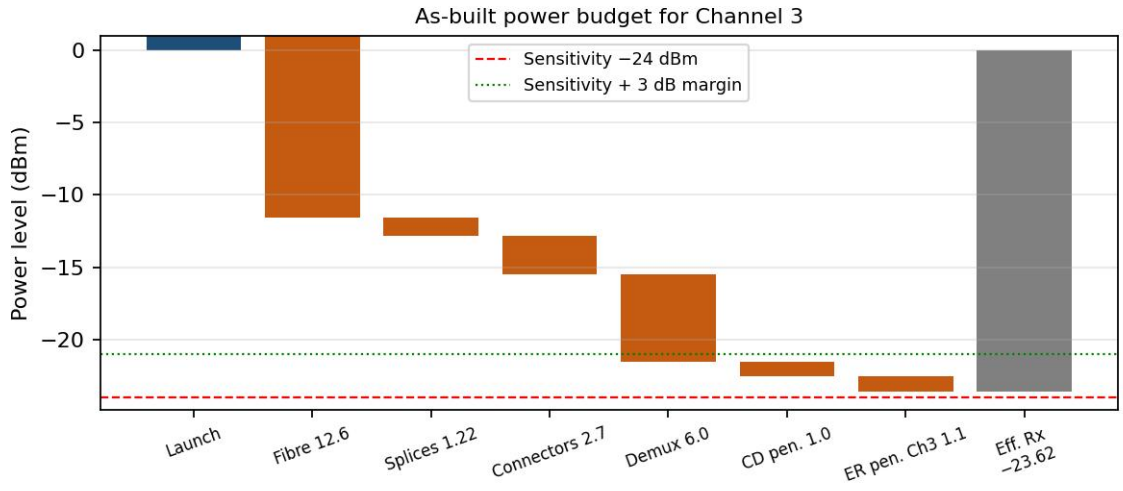


Figure 1: As-built power budget (waterfall) for Channel 3 showing that the effective received power lies only 0.38 dB above sensitivity.

2.3 OSNR

$$OSNR = 58 + P_in - NF = 58 + (-6) - 5 = 47\ dB\ (0.1\ nm)$$

The only amplifier is the booster at the transmitter, and fibre loss after it reduces signal and ASE equally, so the OSNR at the receiver remains ≈ 47 dB. This is 27 dB above the 20 dB requirement. **The link is therefore limited by power and penalties, not by optical noise.**

2.4 PMD

$$\langle \Delta\tau \rangle = 0.1\ ps/\sqrt{km} \times \sqrt{60\ km} = 0.77\ ps\ ;\ tolerance = 0.1 \times T_b = 0.1 \times 100\ ps = 10\ ps$$

The mean DGD is 13 times below the tolerance, so PMD is not a concern at 10 Gb/s.

2.5 Dominant excess loss and correction

Event 3 (+0.70 dB over its 0.5 dB allowance) is the largest single excess, followed by event 2 (+0.57 dB). Repairing both returns the link to the 21.25 dB design loss, (the calculation conservatively keeps the 0.5 dB connector allowance, although a new LC/APC connector is typically 0.25 dB), but the design itself only gives Channel 3 a margin of 1 − 21.25 − 1.00 − 1.10 + 24 = 1.65 dB. The 3 dB margin therefore also needs a launch-power increase. Because OSNR is 47 dB and only 8 channels share the booster, raising the per-channel launch power from +1 to +3 dBm (booster gain 9 dB, total output 12 dBm) is safe from fibre nonlinearity at 10 Gb/s.

Scenario	Loss (dB)	Launch (dBm)	Ch 3 margin (dB)
As-built	22.52	+1	0.38
Repair events 2 and 3 only	21.25	+1	1.65
Repair + launch +2 dBm	21.25	+2	2.65
Repair + launch +3 dBm (selected)	**21.25**	**+3**	**3.65**

DESIGN DECISION 1

I recommend **correcting the excess connector and splice loss first (event 3 then event 2), and then raising the launch power to +3 dBm per channel** because the as-built loss of 22.52 dB leaves Channel 3 with only 0.38 dB margin, the link is power-limited (OSNR 47 dB vs 20 dB required, PMD 0.77 ps vs 10 ps), and the combination of repair (−1.27 dB) and +2 dB launch power raises the margin to 3.65 dB, above the 3 dB requirement.

3. Stage 3 – OTDR, Splice and Connector Configuration

3.1 Pulse width selection

$$One\text{-}way\ trace\ loss \approx 12.60 + 1.22 + 2.70 = 16.52\ dB\ ;\ required\ DR = 16.52 + 6 = 22.52\ dB$$

Pulse	DR (dB)	EDZ / ADZ	DR ≥ 22.5 dB?	Can separate E3/E4 (9 m)?	Can measure E4 loss?
10 ns	20	1 m / 4 m	No	Yes	Yes

Pulse	DR (dB)	EDZ / ADZ	DR ≥ 22.5 dB?	Can separate E3/E4 (9 m)?	Can measure E4 loss?
100 ns	27	5 m / 15 m	Yes (4.5 dB spare)	Yes (9 m > 5 m)	No (9 m < 15 m)
1 µs	33	40 m / 100 m	Yes	No (merged)	No

No single pulse width satisfies everything. The selected **test arrangement** is therefore:

1. **Full-span test, 100 ns, from data centre A**, with a 1 km launch fibre (to move the first connector out of the front-panel dead zone) and a 1 km receive fibre (so that the far-end connector can be measured). This gives enough dynamic range and measures events 1 and 2 and the far-end region.
2. **Short-range test, 10 ns, from data centre B** through the same launch fibre: events 3 and 4 are only ≈ 100 m away, so 20 dB dynamic range is more than enough, and the 4 m ADZ allows the loss of event 4 to be measured.
3. **Bidirectional averaging** (A→B and B→A at 100 ns) for event 1 and event 2, because a one-way OTDR splice reading is affected by the mode-field-diameter difference of the two fibres (giving "gainers" or exaggerated losses).

3.2 Spatial pulse length and resolution

$$v_g = c / n_g = 2.998 \times 10^8 / 1.4682 = 2.042 \times 10^8 \, m/s$$

Pulse width τ	Pulse length in fibre $v_g \cdot \tau$	Two-point resolution $v_g \cdot \tau/2$
10 ns	2.04 m	1.02 m
100 ns	20.4 m	10.2 m
1 µs	204 m	102 m

With 10 ns the 9 m spacing is about nine times the resolution, so events 3 and 4 appear as two distinct reflective spikes with a measurable backscatter level between them. With 100 ns the reflective peaks are still distinguishable (EDZ 5 m), but the backscatter has not recovered after event 3 when event 4 arrives (ADZ 15 m), so event 4's loss is hidden – which is exactly why it is reported as "loss not resolved". With 1 µs both events merge into one large reflection.

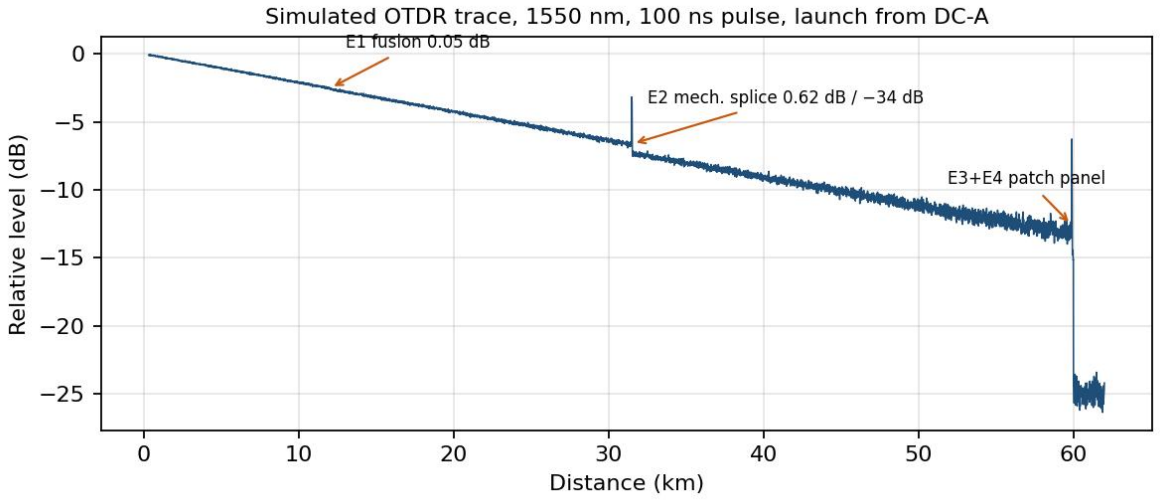


Figure 2: Simulated 100 ns OTDR trace from data centre A showing the four events.

3.3 Event interpretation

Event	Measured	Physical cause	Allowance	Action
1 – 12.00 km	0.05 dB, non-reflective	Good fusion splice	0.05 dB	None
2 – 31.50 km	0.62 dB, −34 dB	Mechanical splice with degraded index-matching gel / fibre end gap or poor cleave; reflection from small air gap	0.05 dB	Replace with fusion splice
3 – 59.90 km	1.20 dB, −22 dB	LC/UPC connector with contaminated, scratched or cracked end face, or air gap (−22 dB is close to the −14.7 dB glass–air Fresnel limit)	0.5 dB	Inspect, clean; replace with LC/APC
4 – 59.909 km	loss not resolved, −40 dB	Second connector of 9 m patch cord – reflectance acceptable for UPC	0.5 dB	Verify with 10 ns + power meter

Fusion vs mechanical splicing: a fusion splice melts the two fibres together with an electric arc, giving 0.02–0.05 dB loss, no air interface (no reflection) and long-term stability. A mechanical splice aligns the fibres in a V-groove with index-matching gel; it is quick and needs no splicer, but loss is typically 0.1–0.5 dB and the gel can dry or move with temperature cycling, which is what the 0.62 dB / −34 dB event shows. For a permanent metro link, fusion splicing is the correct choice.

3.4 Round-trip delay and location uncertainty

$$t_rt = 2\,L\,n_g / c\,;\ \Delta L = L \times \Delta n_g / n_g = L \times 0.002 / 1.4682$$

Event	Distance (km)	Round-trip delay (µs)	Location uncertainty (± m)
1	12.000	117.53	16.3
2	31.500	308.53	42.9
3	59.900	586.69	81.6

Event	Distance (km)	Round-trip delay (µs)	Location uncertainty (± m)
4	59.909	586.78	81.6

A ±0.002 group-index error shifts the absolute position of event 2 by about ±43 m. Before digging at the splice closure, the team should therefore refer to a known landmark (e.g. the cable marker or the 12 km splice closure) rather than relying on the absolute OTDR distance. The relative spacing of events 3 and 4 (9 m) is affected by only ±1 cm.

3.5 Connector selection for the patch panel

LC/APC is selected for the patch-panel repair. LC keeps the same small footprint and is compatible with SFP+ and QSFP-DD pluggables; the 8° angled polish sends reflected light into the cladding, giving return loss ≥ 60 dB (reflectance ≤ −60 dB) with typical insertion loss 0.15–0.25 dB. SC would require new adapters and more space; MPO is for parallel multi-fibre optics and has higher loss (0.35–0.75 dB); CS is intended for next-generation 400G/800G module front panels and is not needed at the patch panel. The equipment-side ports remain UPC, so hybrid LC/APC-to-LC/UPC patch cords are used; APC and UPC must never be mated directly, as the angle mismatch causes several dB of loss.

3.6 Repair priority and verification

1. **Priority 1 – event 3 (patch panel, DC-B):** largest excess loss (+0.70 dB), worst reflectance (−22 dB, which also causes multipath interference and laser instability), located indoors and repairable in minutes. Inspect with a fibre microscope to IEC 61300-3-35, clean, and if damaged replace with an LC/APC connector.
2. **Priority 2 – event 2 (field splice closure, 31.5 km):** +0.57 dB excess; requires a splicing crew to open the closure and replace the mechanical splice with a fusion splice.
3. **Priority 3 – event 4:** check with the 10 ns test from DC-B and the power meter.

Verification: after each repair, (i) repeat the 100 ns bidirectional OTDR test – event 2 should drop to ≤ 0.05 dB with no reflection, event 3 to ≤ 0.5 dB with reflectance ≤ −55 dB; (ii) measure end-to-end insertion loss with light source and power meter (one-cord reference method, IEC 61280-4-2) – target ≤ 21.25 dB.

DESIGN DECISION 2

I recommend **giving event 3 (the 1.20 dB, −22 dB LC/UPC patch-panel connector) the highest repair priority, replacing it with an LC/APC connector, then fusion-splicing event 2, with the OTDR used at 100 ns bidirectionally for the full span plus a 10 ns test from data centre B with launch and receive fibres** because event 3 has the largest excess loss and the worst reflectance, is the easiest to reach, and only a 10 ns pulse (ADZ 4 m) can resolve the loss of event 4, which is just 9 m away, while 100 ns provides the 22.5 dB dynamic range needed for the full 60 km span.

4. Stage 4 – Eye Diagram, BER and Quantum Limit

4.1 BER from Q

$$BER = \tfrac{1}{2}\,\mathrm{erfc}(Q/\sqrt{2})$$

Channel	Q	BER	vs 10^{-12}
3	7.6	1.5×10^{-14}	Passes (≈ 70× better)
5	4.7	1.3×10^{-6}	Fails (≈ 10^6× worse)
Target	7.03	1.0×10^{-12}	–

4.2 Jitter and eye width

$$TJ(10^{-12}) = DJ + 14.07 \times RJ$$

Channel	DJ (UI p-p)	RJ (UI rms)	TJ (UI)	Eye width = 1 − TJ (UI)
3	0.18	0.012	0.18 + 0.169 = 0.349	0.651
5	0.30	0.022	0.30 + 0.310 = 0.610	0.390

Channel 5 has lost 40% of its horizontal eye compared with Channel 3. The **vertical** closure (noise margin) shows up in Q, while the **horizontal** closure (timing margin) shows up in TJ. Channel 5 is worse on both axes, which points to its transmitter (laser driver, bias, clock data recovery at the source), since fibre-induced jitter at 1 ps PMD would be negligible.

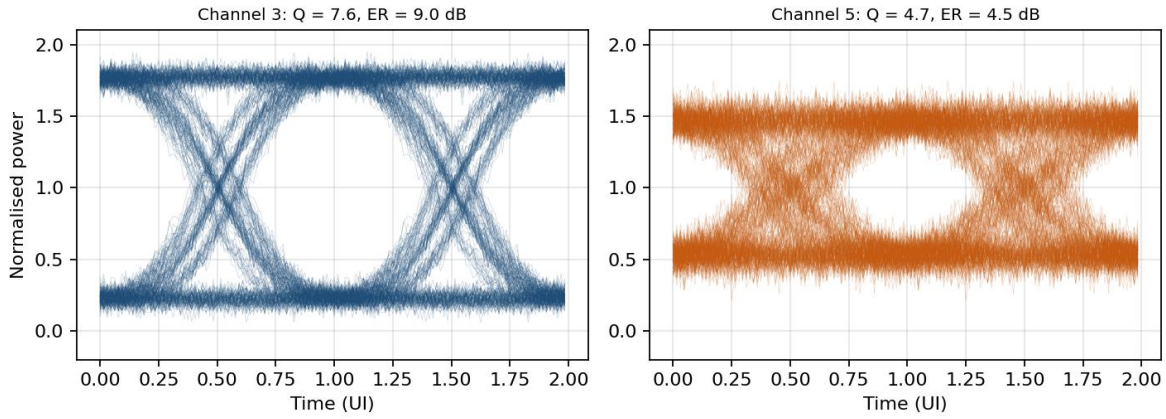


Figure 3: Simulated eye diagrams for Channel 3 (open) and Channel 5 (reduced ER, larger jitter).

4.3 Root cause: shared or channel-specific?

At the same average received power, Q is proportional to (r − 1)/(r + 1):

$$Q_5\ (predicted) = Q_3 \times [(r_5 - 1)/(r_5 + 1)] / [(r_3 - 1)/(r_3 + 1)] = 7.6 \times 0.476 / 0.776 = 4.66$$

The predicted value 4.66 agrees almost exactly with the measured 4.7. In optical dB, $10\log_{10}(7.6/4.7) = 2.09$ dB, while the ER-penalty difference is 3.22 − 1.10 = 2.12 dB. **The extinction ratio of Channel 5 fully explains its Q deficit.** Both channels share the same fibre, connectors, splices and dispersion (the 100 GHz difference in wavelength changes D by less than 0.1%), so a fibre fault cannot produce such a large difference. The root cause of the

Channel 5 eye closure is its transmitter: incorrect laser bias (0 level too high) or insufficient modulation drive, plus a noisy driver adding DJ and RJ. The fibre faults (events 2 and 3) are the shared cause of the low margin on all channels.

4.4 Quantum limit

$$BER = \tfrac{1}{2}\exp(-N_1) = 10^{-12} \Rightarrow N_1 = \ln(5 \times 10^{11}) = 26.9 \text{ photons per "1" bit}$$

With equal numbers of 1s and 0s the average is $\bar{N}$ = 13.5 photons/bit.

$$h\nu = hc/\lambda = (6.626 \times 10^{-34})(2.998 \times 10^{8})/(1550 \times 10^{-9}) = 1.282 \times 10^{-19}\,J$$

$$P_QL = \bar{N} \cdot h\nu \cdot B = 13.5 \times 1.282 \times 10^{-19} \times 10^{10} = 1.73 \times 10^{-8}\,W = -47.6\,dBm$$

The practical receiver (−24 dBm) is therefore **23.6 dB above the quantum limit**. This large gap is normal for a PIN receiver, whose sensitivity is set by thermal noise of the amplifier rather than by photon statistics.

4.5 Coherent detection comparison

Scheme	BER expression	Photons/bit at 10^{-12}	Sensitivity at 10 Gb/s
Ideal direct detection (OOK)	½ exp(−N_1)	13.5 (avg), 26.9 per 1-bit	−47.6 dBm
Homodyne PSK	½ erfc(√(2N))	12.4	−48.0 dBm
Heterodyne PSK	½ erfc(√N)	24.7	−45.0 dBm
Practical PIN receiver	–	≈ 3.1 × 10^3 (equivalent)	−24 dBm

For homodyne PSK, ½ erfc(x) = 10^{-12} gives x = 4.974, so 2N = 24.7 and N = 12.4 photons/bit; heterodyne needs twice as many (24.7) because the IF stage adds image-band noise.

4.6 Implications for the upgrade

A coherent receiver mixes the weak signal with a strong local oscillator, so the LO shot noise dominates the thermal noise and the receiver approaches the shot-noise limit within a few dB. This is why coherent DP-QPSK/16-QAM modules can work at OSNR values of 12.5–26 dB and receive powers far below what a direct-detection receiver would need at the same bit rate. On this repaired link with OSNR ≈ 47 dB, all three coherent options are noise-safe, and the choice can be made on spectrum and energy instead.

DESIGN DECISION 3

I recommend **declaring Channel 5 not acceptable and correcting its transmitter (reset laser bias and modulator drive, or replace the 10G DWDM SFP+) until ER ≥ 9 dB, then confirming recovery by Q ≥ 7.03 (target ≈ 7.6), TJ ≤ 0.35 UI and BER ≤ 10^{-12} on the BERT** because its measured BER (1.3 × 10^{-6}) is a million times worse than the target, its 4.5 dB ER alone predicts Q = 4.66 (matching the measured 4.7), and its 0.39 UI eye width shows transmitter jitter – while the receiver itself is still 23.6 dB away from the quantum limit and is not the cause.

5. Stage 5 – Integrated WDM Upgrade and Energy Strategy

5.1 G.694.1 grid

$$f_n = 193.1\ THz + n \times 0.05\ THz;\ n = +3 \Rightarrow f = 193.25\ THz$$

$$\lambda = c/f = 2.998 \times 10^8 / 193.25 \times 10^{12} = 1551.36\ nm$$

Option Q slot count: occupied bandwidth = $R_s(1 + \beta) = 32 \times 1.1 = 35.2$ GHz → $\lceil 35.2/12.5 \rceil = 3$ slots = 37.5 GHz per channel.

5.2 Spectrum, efficiency and OSNR margin

Option	Configuration	Occupied spectrum	Payload SE (b/s/Hz)	Required OSNR	OSNR margin (47 dB link)
P	16 × 100G DP-QPSK, 50 GHz	16 × 50 = 800 GHz	100/50 = 2.0	12.5 dB	34.5 dB
Q	8 × 200G DP-16QAM, 37.5 GHz	8 × 37.5 = 300 GHz	200/37.5 = 5.33	19.5 dB	27.5 dB
R	4 × 400G, 75 GHz	4 × 75 = 300 GHz	400/75 = 5.33	26 dB	21.0 dB

5.3 Energy per bit

$$E_{bit} = (N_{modules} \times P_{module}) / 1.6\ Tb/s$$

Solution	Modules for 1.6 Tb/s	Total power (W)	Energy per bit (pJ/bit)
Legacy 10G DWDM SFP+	160 × 3.5 W	560	350
P – 100G DP-QPSK	16 × 22 W	352	220
Q – 200G DP-16QAM	8 × 20 W	160	100
R – 400G Si-photonic	**4 × 15 W**	**60**	**37.5**

Silicon photonics integrates the IQ modulators, polarisation splitters/rotators, 90° hybrids and germanium photodiodes on one CMOS-compatible chip, co-packaged with a 7 nm (or smaller) DSP. This removes discrete optical sub-assemblies, shortens RF interconnects and allows the whole coherent transceiver to fit in a QSFP-DD/OSFP pluggable that plugs directly into a router port. The result is the factor-of-9 energy saving over legacy 10G optics shown in Figure 4.

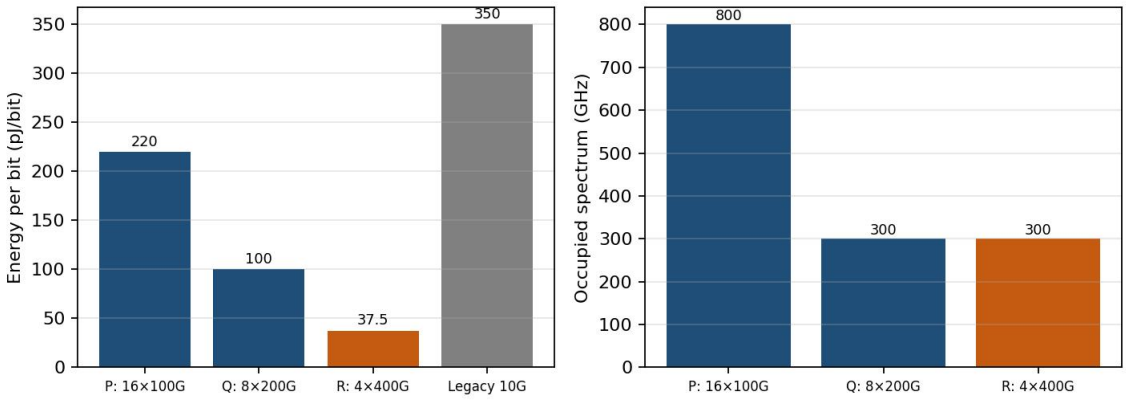


Figure 4: Energy per bit and occupied spectrum of the upgrade options.

5.4 Channel plan for Option R

Each 400G channel occupies a 75 GHz flex-grid slot (m = 6 × 12.5 GHz). The existing 100 GHz demultiplexer cannot pass 75 GHz-spaced channels, so it is replaced by a flex-grid thin-film mux/demux (≈ 3.5 dB loss).

Channel	Centre frequency (THz)	Wavelength (nm)	Slot
R1	193.100	1552.56	n = 0, m = 6
R2	193.175	1551.96	n = 12, m = 6
R3	193.250	1551.36	n = 24, m = 6
R4	193.325	1550.76	n = 36, m = 6

Power check: link loss with new mux = 12.60 + 0.65 + 2.00 + 3.5 = 18.75 dB; with four channels sharing the booster's original 10 dBm total output, each channel can be launched at +4 dBm, so the received power is ≈ −14.8 dBm per channel – comfortably inside the input range of coherent pluggables.

5.5 Sequence of work

1. Repair event 3 and event 2; verify with OTDR and power meter (loss ≤ 21.25 dB).
2. Correct Channel 5 transmitter; verify all 8 legacy channels at Q ≥ 7.03 so that existing traffic is safe during migration.
3. Install flex-grid mux/demux and the 400G pluggables; migrate traffic channel by channel.
4. Repeat OSA (OSNR ≥ 26 dB + margin), pre-FEC BER from module DSP, and end-to-end loss.

Repairing before upgrading is essential: a −22 dB reflection at the patch panel would also degrade the coherent receivers through multipath interference, and the extra 1.27 dB of loss would be carried into the new design.

5.6 Ranking

Criterion	P	Q	R

Criterion	P	Q	R
Spectrum used	3rd (800 GHz)	1st = (300 GHz)	1st = (300 GHz)
OSNR margin	1st (34.5 dB)	2nd (27.5 dB)	3rd (21.0 dB – still large)
Energy per bit	3rd (220 pJ)	2nd (100 pJ)	1st (37.5 pJ)
Number of modules	16	8	4

DESIGN DECISION 4

I recommend **the integrated workflow "repair → verify → fix Channel 5 → upgrade to Option R (4 × 400G Si-photonic pluggables in 75 GHz flex-grid slots)"** because it delivers 1.6 Tb/s on the existing fibre pair in only 300 GHz (5.33 b/s/Hz), keeps a 21 dB OSNR margin on the repaired 47 dB OSNR link, and uses only 37.5 pJ/bit – 2.7× less than Option Q and 9.3× less than the legacy 10G optics.

6. Stage 6 – Verification and Final Design

6.1 Verification table

Requirement	Specified / Target	Achieved / Evidence
Total optical link loss	≤ 21.25 dB (design)	As-built 22.52 dB; after repair 21.25 dB (18.75 dB with new flex-grid mux for upgrade)
Received optical power	≥ −24 dBm plus 3 dB margin	After repair and +3 dBm launch: −18.25 dBm; effective −20.35 dBm (Ch 3)
System margin	≥ 3 dB	0.38 dB (as-built) → 3.65 dB (repaired, +3 dBm)
OTDR event condition	No event above allowance	E2: 0.62 → ≤ 0.05 dB (fusion); E3: 1.20 → ≤ 0.25 dB (LC/APC), reflectance ≤ −60 dB
Eye / Q / jitter / ER	Q ≥ 7.03; ER ≥ 8.2 dB	Ch 3: Q 7.6, ER 9 dB, TJ 0.35 UI. Ch 5: Q 4.7, ER 4.5 dB → after Tx fix ER ≥ 9 dB, Q ≈ 7.6
BER and quantum-limit gap	BER ≤ 10^{-12}	Ch 3: 1.5 × 10^{-14}; Ch 5: 1.3 × 10^{-6} → ≤ 10^{-12} after fix; Rx 23.6 dB above quantum limit (−47.6 dBm)
Upgrade spectrum and energy per bit	1.6 Tb/s, minimum pJ/bit	Option R: 4 × 400G, 300 GHz, 5.33 b/s/Hz, 37.5 pJ/bit, OSNR margin 21 dB

The results are consistent with the original symptoms: the power meter showed low margin on all channels (shared cause = events 2 and 3), while only Channel 5 showed a closed eye (channel-specific cause = its transmitter ER). The high OSNR confirms that neither symptom was caused by noise.

6.2 Diagnostic and upgrade workflow

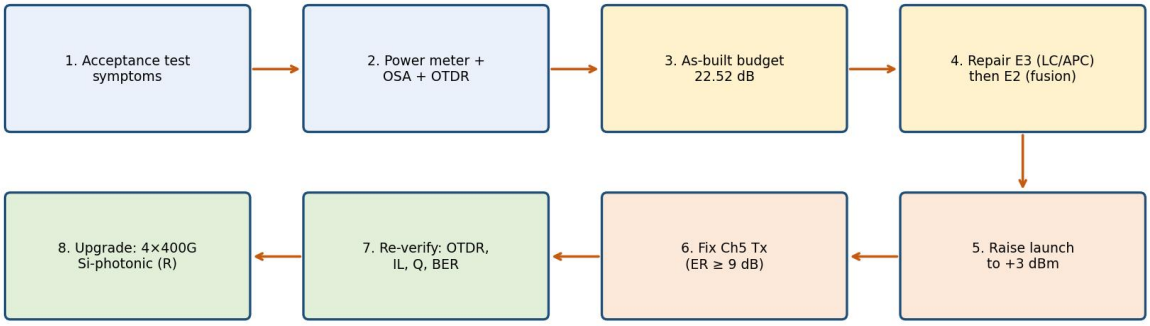


Figure 5: Diagnostic, repair and upgrade workflow adopted for the 60 km link.

6.3 Final parameter and decision table

Item	Value / decision adopted
Fibre	60 km G.652.D, 0.21 dB/km, n_g 1.4682 – retained
Corrected link loss	21.25 dB (legacy), 18.75 dB (upgrade with flex-grid mux)
Launch power	+3 dBm/ch (legacy 8 ch); +4 dBm/ch (4 × 400G)
CD penalty / PMD	1.00 dB / 0.77 ps (≪ 10 ps)
OSNR	47 dB (booster-only link)
OTDR settings	100 ns bidirectional full span + 10 ns from DC-B; 1 km launch and receive fibres
Repair priority	1) E3 → LC/APC 2) E2 → fusion splice 3) verify E4
Channel 5	Transmitter bias/drive correction or replacement, ER ≥ 9 dB
Quantum limit	−47.6 dBm (DD), −48.0 dBm (homodyne), −45.0 dBm (heterodyne)
Upgrade	Option R, 4 × 400G at 193.100–193.325 THz, 75 GHz slots
Energy	60 W per end, 37.5 pJ/bit

7. Final Engineering Decision

YES – I would deploy the optimised design. Justification:

- **Link budget and margin:** repair plus +3 dBm launch gives 3.65 dB margin after the 1.0 dB CD penalty and 1.1 dB ER penalty – above the 3 dB requirement.
- **Total loss and dominant excess:** the 1.27 dB excess was traced to event 3 (0.70 dB) and event 2 (0.57 dB); both are removed and the link returns to its 21.25 dB design value.
- **OTDR diagnosis:** the 100 ns/10 ns two-pulse strategy overcomes the dead-zone limit at the 9 m-spaced patch panel; repair priority follows loss, reflectance and accessibility.
- **Eye, jitter, ER, Q, BER:** Channel 3 already meets 10^{-12}; Channel 5 is a transmitter fault whose ER fully explains its Q, and is corrected by transmitter adjustment/replacement.

- **Quantum limit and coherent detection:** the direct-detection receiver is 23.6 dB from the quantum limit; coherent receivers approach the shot-noise limit and are well suited to the 47 dB OSNR link.
- **Channel plan, SE and energy:** Option R on the G.694.1 flex-grid uses 300 GHz at 5.33 b/s/Hz and 37.5 pJ/bit.
- **Overall suitability:** 1.6 Tb/s can be carried reliably over the 60 km link without re-laying fibre.

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